ILLINOIS

Air Conditioning and
Refrigeration Center

Water-Cooled Chiller for High Efficiency Edge Compute Data Center Facility

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Course Project: IRCC 2026 — Project #3

150 kW water-cooled propane chiller · Champaign, IL

Cycle design, component selection & annual performance

Introduction

Compute Demand Is Climbing — and Cooling Has to Keep Up

- **Compute demand is exploding** — more racks, more heat
- Cooling is the **largest non-IT load** — it decides **PUE**
- Runs **24/7/365** — cooling can never stop
- **Every watt saved on cooling is a watt for compute**

Image: gridx.com

The Design Task

Deliverables & Hard Targets

- Full **cycle design** with working-fluid comparison
- **Selection & sizing** of every major component
- Seasonal performance — **COP, IPLV, PUE**
- **BOM · p–h chart · P&ID** · safety · controls

PARAMETER	REQUIREMENT
Maximum cooling load	150 kW (≈42 RT), sensible IT
Minimum load / turndown	30 kW — 5:1
Chilled water supply / return	15 / 21 °C (ASHRAE TC 9.9 W17)
White space	8 × 12 m · 25 racks · 24/7/365
Design ambient	35 °C DB · TMY3 Champaign
Full-load COP	≥ 3.5 at the AHRI point (ASHRAE 90.1)
IPLV.IP	≥ 5.0 (AHRI 550/590-2023)
Economizer	active ≤ 10 °C · full ≤ 4 °C OAT (§6.5.1)
Refrigerant	GWP < 150 · A2L/A3 + charge analysis

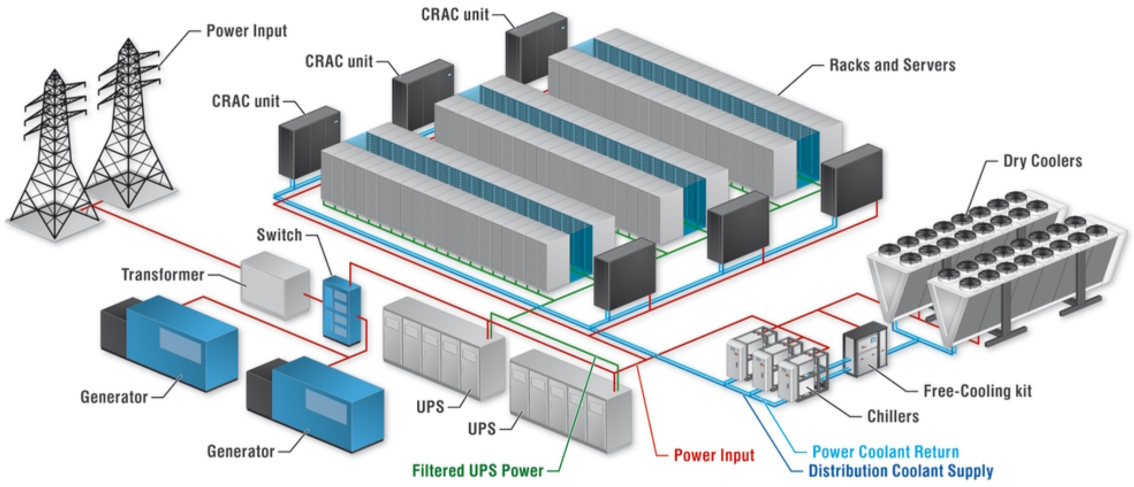
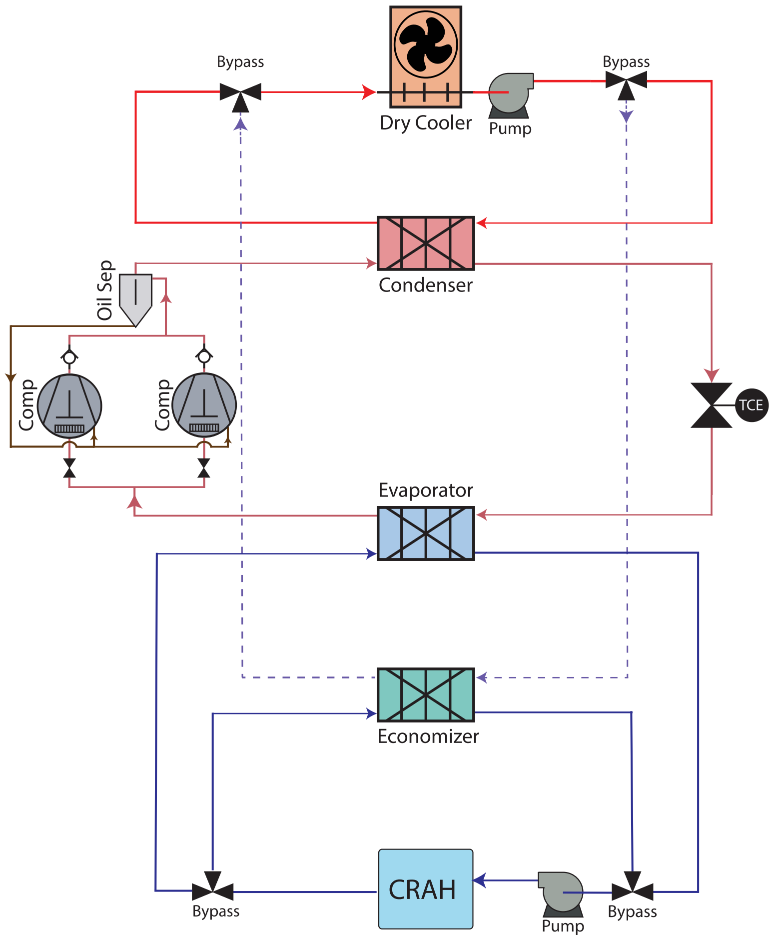


Image: cooltec-systems.com

Proposed Design

Three Loops in Series, Economizer in Parallel

- Warm **chilled water** carries the chips' heat back from the **CRAH units**
- Warm water routes to the **economizer** (free cooling) or the chiller's **evaporator**
- In the **evaporator** the **R-290 loop** absorbs the heat; the **compressor** lifts it
- The **condenser** passes the heat to the **glycol loop** → the **dry cooler** rejects it outdoors

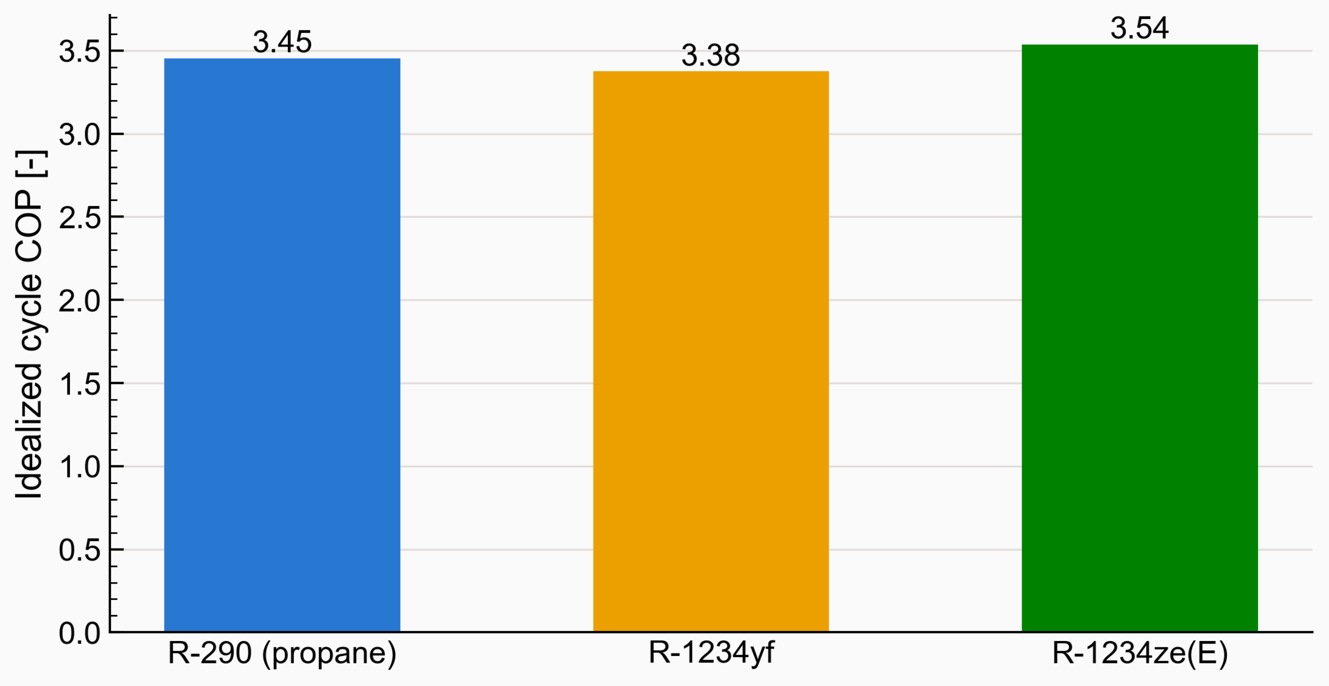


System P&ID — refrigerant, glycol and chilled-water loops

Working-Fluid Selection

COP of the Screened Low-GWP Candidates

- **1.7× more cooling per swept volume** → smaller compressor
- Rivals win **~2% COP** — but need a bigger machine
- **Bigger, costlier, bulkier** — not worth ~2%



Cycle COP across screened refrigerant candidates

Idealized single-stage cycle · 7 / 52 °C · 8 K superheat · 5 K subcool · η_{isen} = 0.70

SELECTED WORKING FLUID

R-290

- GWP 3
- Zero ODP
- High volumetric capacity
- Low cost
- Widely available

A future-proof natural refrigerant

Evaporator — Brazed-Plate HX

Evaporator — two-zone model

- **Two zones:** boiling + superheat, split by enthalpy
- 150 kW → 8.2 m² at **U = 2,400 W/m²K** → **≈84 plates** (~20% margin)
- **Flow parameters stay inside the correlation's validity range**

ΔP — REFRIGERANT SIDE (BOILING ZONE)

$$\Delta P_{\mathrm{fr}} = f \cdot (L_v\, N_{\mathrm{cp}} / D_h) \cdot G_{\mathrm{Eq}}^2 / \rho_f$$

$$f = Ge_3 \cdot Re_{\mathrm{Eq}}^{Ge_4} \qquad G_{\mathrm{Eq}} = G_c\left[(1-x) + x(\rho_f/\rho_g)^{1/2}\right] \qquad G_c = \dot{m}/(N_{\mathrm{cp}}\, b\, L_w)$$

ΔP — WATER SIDE

$\Delta P = 2f \cdot (L_v / D_h) \cdot G^2/\rho$ — Martin (1996), single-phase

5.973 kg/s · G = 545 kg/m²s · Re = 2,072 · f = 0.109 → **ΔP = 9.75 kPa**

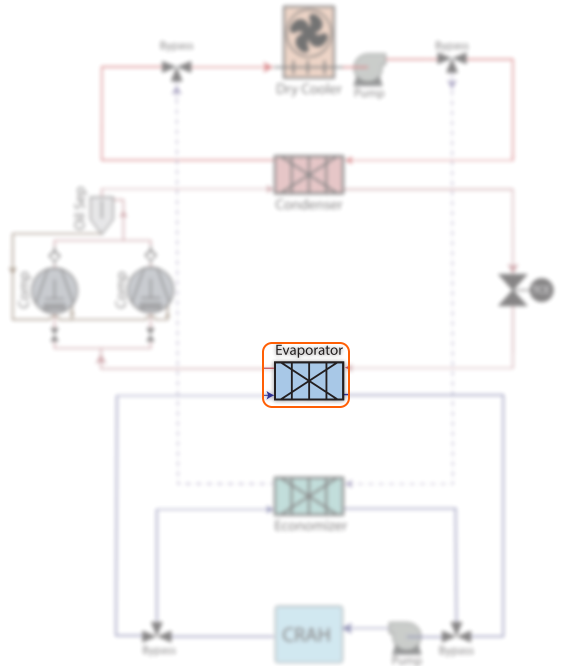
WHERE F COMES FROM — HAN (2003), EVAPORATION SET

$$Ge_3 = 64{,}710 \cdot (p_{\mathrm{co}}/D_h)^{-5.27} \cdot (\pi/2 - \beta)^{-3.03}$$

At β = 30°, p_{co} = 5 mm, D_h = 4 mm → **Ge_3 ≈ 17,361, Ge_4 ≈ −1.120**

D_h 4 mm · β 30° · p_{co} 5 mm · b 2.34 mm (derived) · L_w 200 mm · L_v 0.5 m — from HX geometry

SELECTED MODEL → **Kelvion HP DW 500H · double-wall · ≈84 plates**



Evaporator — location in the system

Kelvion double-wall brazed-plate HX

Condenser — Brazed-Plate HX

Condenser — three-zone model

- **Three zones** by enthalpy — 14 / **81.5** / 4.5%; one LMTD suffices
- 182 kW at **t_c = 45 °C** → 12.2 m² at **U = 2,000 W/m²K** → **≈104 plates**
- **5 K subcool** held in the condenser → no receiver

ΔP — REFRIGERANT SIDE (CONDENSING ZONE)

$\Delta P_{\mathrm{fr}} = f \cdot (L_v\, N / D_h) \cdot G_{\mathrm{Eq}}^2 / \rho_f$ — integrated over 200 quality steps, x = 0 → 1 (np.trapezoid)

Desuperheat + subcool zones (Blasius): $\Delta P = 2f \cdot (G^2/\rho) \cdot L_{\mathrm{zone}} / D_h$

N = 2 is Han's rig constant, not N_{cp} = 47 (else 23.5× over) · G_c ≈ 25.1 — inside Han's 13–34 band

ΔP — GLYCOL SIDE

$\Delta P = 2f \cdot (L_v / D_h) \cdot G^2 / \rho$ — Martin (1996)

11.64 kg/s · Re = 2,703 (turbulent branch) · f = 0.107 → **ΔP = 35.91 kPa**

4× the evaporator water loss at equal area — ~2× flow, denser fluid, ΔP ∝ G²

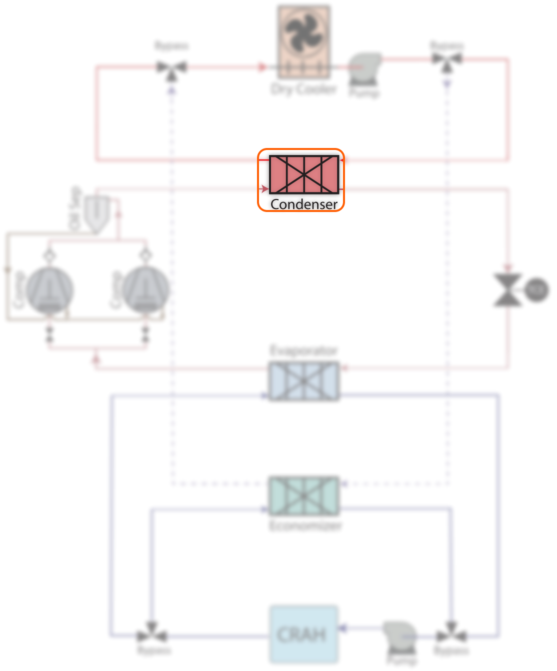
WHERE F COMES FROM — HAN (2003), CONDENSATION SET

$Ge_3 = 3{,}521.1 \cdot (p_{\mathrm{co}} / D_h)^{+4.17} \cdot (\pi/2 - \beta)^{-7.75}$

Same β and p_{co} as the evaporator → **Ge_3 ≈ 6,246, Ge_4 ≈ −0.985**

Sign flips (+4.17 vs −5.27): film growth vs nucleation — **f ≈ 2.8× lower**

SELECTED MODEL → **Kelvion HP DW 500H · double-wall · ≈104 plates**



Condenser — location in the system

Kelvion double-wall brazed-plate HX

Economizer Selection and Modeling

- Sized for full **150 kW at 4 °C OAT**
- **≈49 plates** · single-wall — **no refrigerant contact**
- **Free cooling — 40% of the year**

ΔP — BOTH SIDES, SAME EQUATION

$$\Delta P = 2f \cdot (L_v/D_h) \cdot G^2/\rho \qquad G = \dot{m}/(N_{\mathrm{cp}}\, b\, L_w)$$

Water: Re = 1,617 · f = 0.105 → **6.67 kPa** | Glycol: Re = 456 · f = 0.142 → **10.43 kPa**

Both on the laminar branch — the condenser's glycol at Re = 2,703 is turbulent

WHERE F COMES FROM — MARTIN (1996)

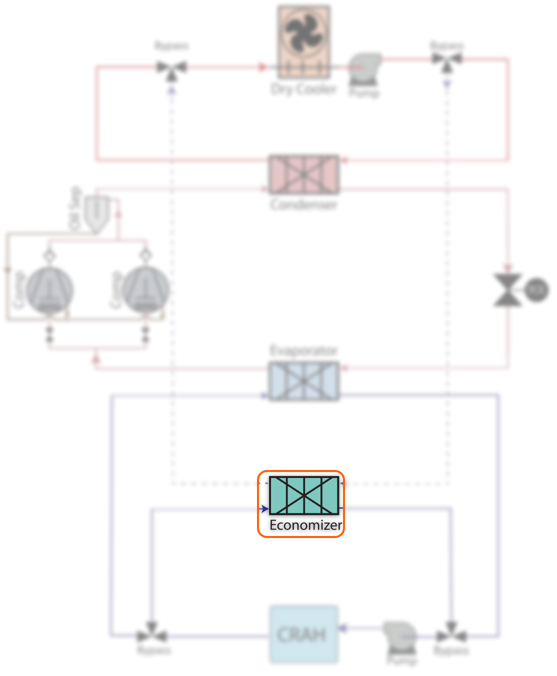
$$\frac{1}{\sqrt{f}} = \frac{\cos\varphi}{\sqrt{0.045\tan\varphi + 0.09\sin\varphi + f_0/\cos\varphi}} + \frac{1-\cos\varphi}{\sqrt{3.8\, f_1}}$$

$$Re < 2000: \quad f_0 = 16/Re, \quad f_1 = 149/Re + 0.9625$$

φ is the chevron angle, used directly as β = 30° — no 90°−β conversion

One shared _martin_dp() serves every single-phase stream in the plant

SELECTED MODEL ⟶ **Kelvion GB series · single-wall · ≈49 plates**



Economizer — location in the system

Kelvion plate heat exchanger, glycol → chilled water

Compressor Selection and Modeling

- **Parallel pair** for turndown: one compressor covers the **30 kW minimum** efficiently
- Modeled on the **AHRI-540 polynomial map** — the manufacturer's own data
- **2 × ≈75 kW** Bitzer 4FEP-35Z, **parallel** on one shared circuit

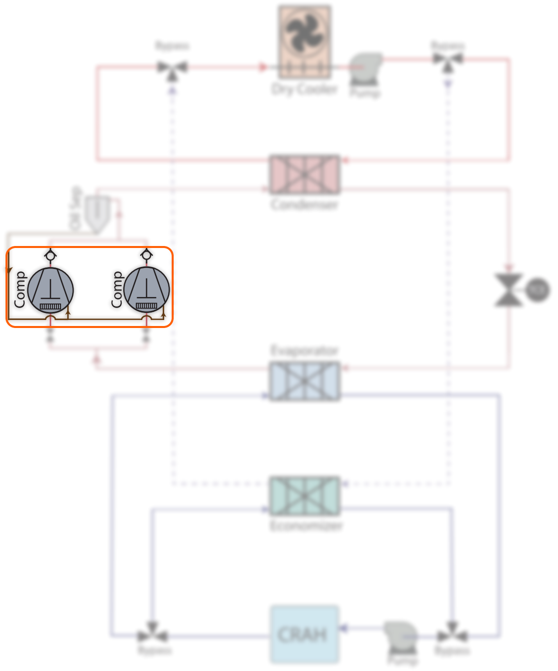
$$y = c_1 + c_2 t_o + c_3 t_c + c_4 t_o^2 + c_5 t_o t_c + c_6 t_c^2 + c_7 t_o^3 + c_8 t_c t_o^2 + c_9 t_o t_c^2 + c_{10} t_c^3$$

AHRI-540 / EN 12900 — separate coefficient sets for $\dot{Q}$, P and $\dot{m}$

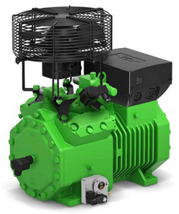
$$\dot{Q}(N) = (N/N_0) \cdot \dot{Q}_0 \qquad P(N) = (N/N_0) \cdot P_0$$

VFD model — capacity and power scale ≈linearly with speed for a recip, so COP is speed-invariant

SELECTED MODEL ⟶ **2 × Bitzer 4FEP-35Z · semi-hermetic recip · VFD**



Compressors — location in the system



Bitzer ECOLINE semi-hermetic reciprocating, R-290-rated

Electronic Expansion Valve

Expansion Valve Selection and Modeling

- **One EEV** sized for the **full 150 kW** load
- **82.6% open at full load → 24.3% at 30 kW** across the **5:1 turndown**, above the **~15% floor**
- Stays **controllable across the whole range** — confirmed by the **manufacturer's data**

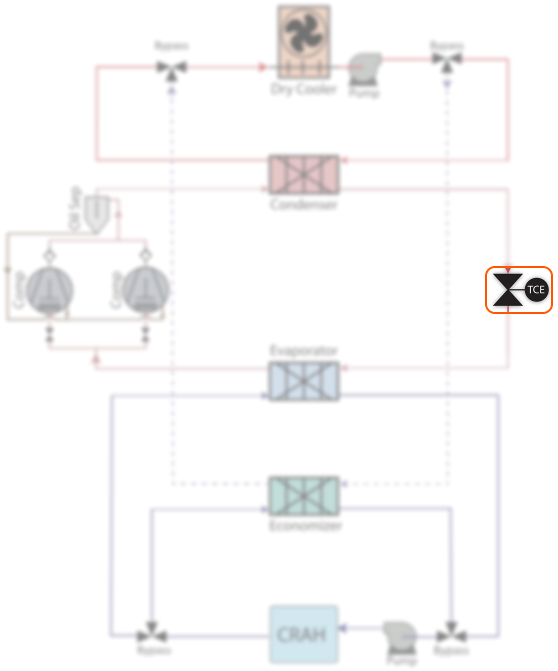
$$\dot{m}_r = C_d \cdot A_{\text{orifice}} \cdot \sqrt{2\rho_{\text{in}}\,\Delta P} \quad \Rightarrow \quad A_{\text{orifice}} = \frac{\dot{m}_r}{C_d\sqrt{2\rho_{\text{in}}\,\Delta P}}$$

Incompressible-liquid orifice equation, inverted for the required flow area

$\text{opening} = A_{\text{req}}(\text{load})/A_{\text{orifice}}$ → 82.6% at 150 kW · 24.3% at 30 kW

Valve-authority check across the 5:1 turndown (controls chapter)

SELECTED MODEL ⟶ **Emerson EX class · single valve**



Expansion valve — location in the system

Electronic expansion valve, stepper-driven

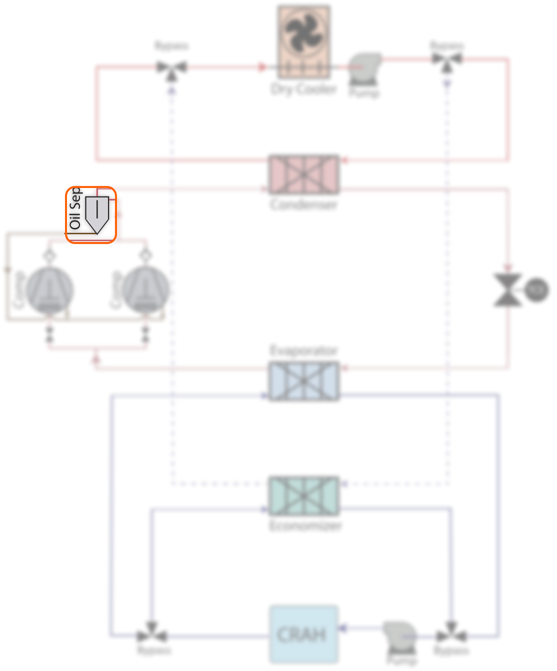
Oil Separator

Oil Separator Selection and Modeling

- On the **discharge outlet** · returns oil to the **compressor sump**
- Selected on the **peak volume flow** from the compressor map
- **Bitzer** unit — matched oil, **single-vendor** with the compressors

> $\dot{V}_{\text{suction}} = \dot{m}_r / \rho_{\text{suction}}$ · 2 units → **203.6 m³/h** → OA1954
>
> Bitzer quick-selection chart: sized on combined suction volume flow

SELECTED MODEL → **Bitzer OA1954 · primary · A3-rated · 28 bar**



Oil separator — location in the system

Bitzer OA-series primary oil separator

Dry Cooler

Dry Cooler Selection and Modeling

- Rejects **≈196 kW** (670 MBH) to ambient
- **Fan power scales as N^3** (cube law) — modeled hour-by-hour in the annual sim
- **ΔP from vendor-anchored scaling** — Kelvion's rated point via the **Blasius ratio**, geometry cancels

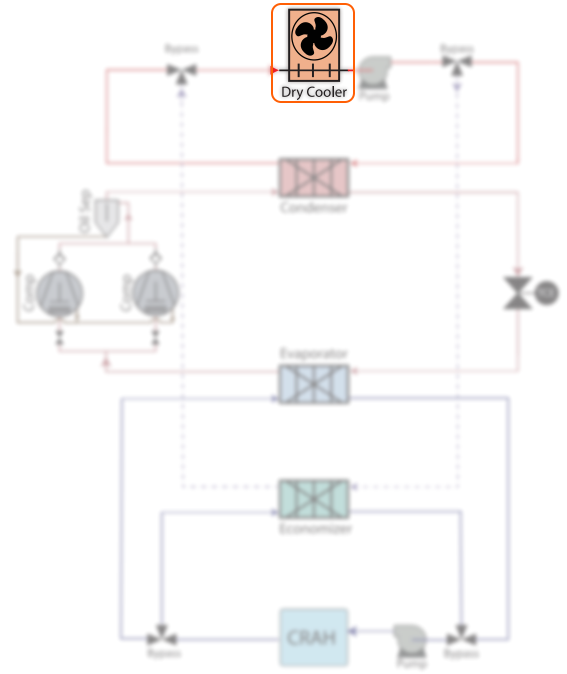
$$\varepsilon = 1 - \exp\left\{(1/C_r)\, NTU^{0.22}\left[\exp(-C_r\, NTU^{0.78}) - 1\right]\right\} \qquad UA = NTU \cdot C_{\min}$$

Cross-flow ε-NTU (both streams unmixed) — NTU found by numerical root-finding

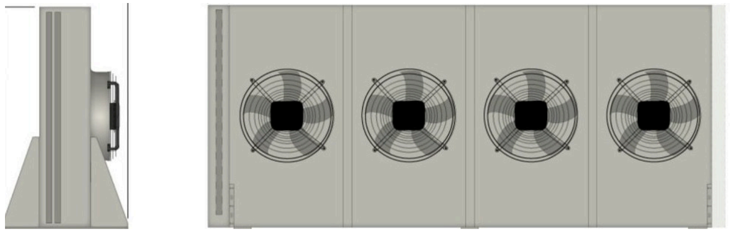
$$\Delta P_2/\Delta P_1 = (\rho_2/\rho_1)^{0.8}\,(\dot{m}_2\rho_1/\dot{m}_1\rho_2)^{1.8}\,(\mu_2/\mu_1)^{0.2}$$

Vendor-anchored ΔP scaling — geometry cancels; needs one rated point + CoolProp only

SELECTED MODEL → **Kelvion ULF-PA106K4V-091F095 · 6 EC fans**



Dry cooler — location in the system



Kelvion flat dry cooler, variable-speed EC fans

Circulation Pumps

Pump Selection and Modeling

- Pumps **sized to deliver the loop flow** against its required **head**
- **Digitized curves** → pump power in the PUE model
- **B&G e-1510** · **2.5AC** (glycol) + **1.5AD** (CHW)

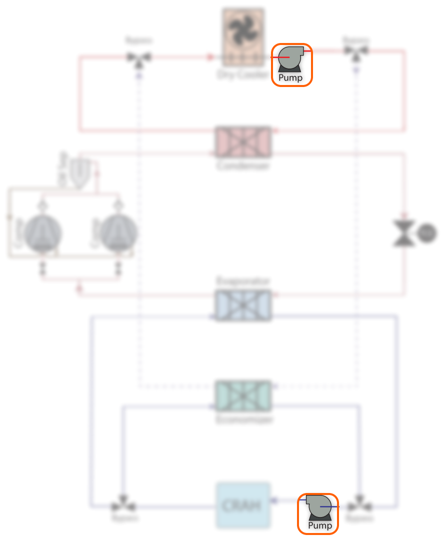
$$H(Q) = c_2 Q^2 + c_1 Q + c_0 \qquad P_{\text{shaft}} = \rho\, g\, Q\, H/\eta_{\text{pump}}$$

Quadratic fit to each digitized trim curve · trim chosen where H(Q) clears the margined duty

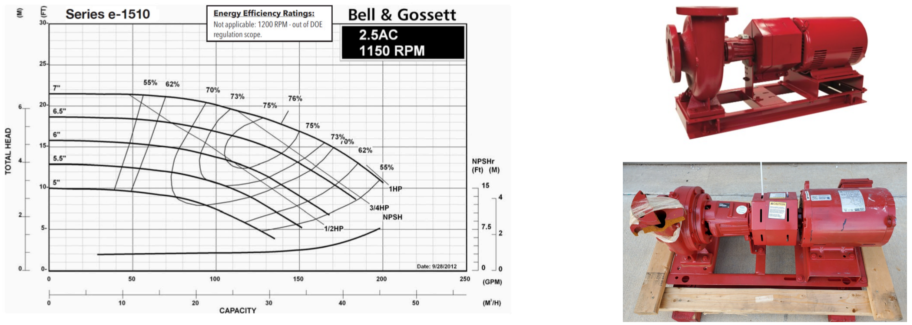
$$Q \propto N \qquad H \propto N^2 \qquad P \propto N^3$$

Affinity laws — VFD part-load power in the annual model

SELECTED MODEL → **Bell & Gossett e-1510 · both VFD · 1750 RPM**



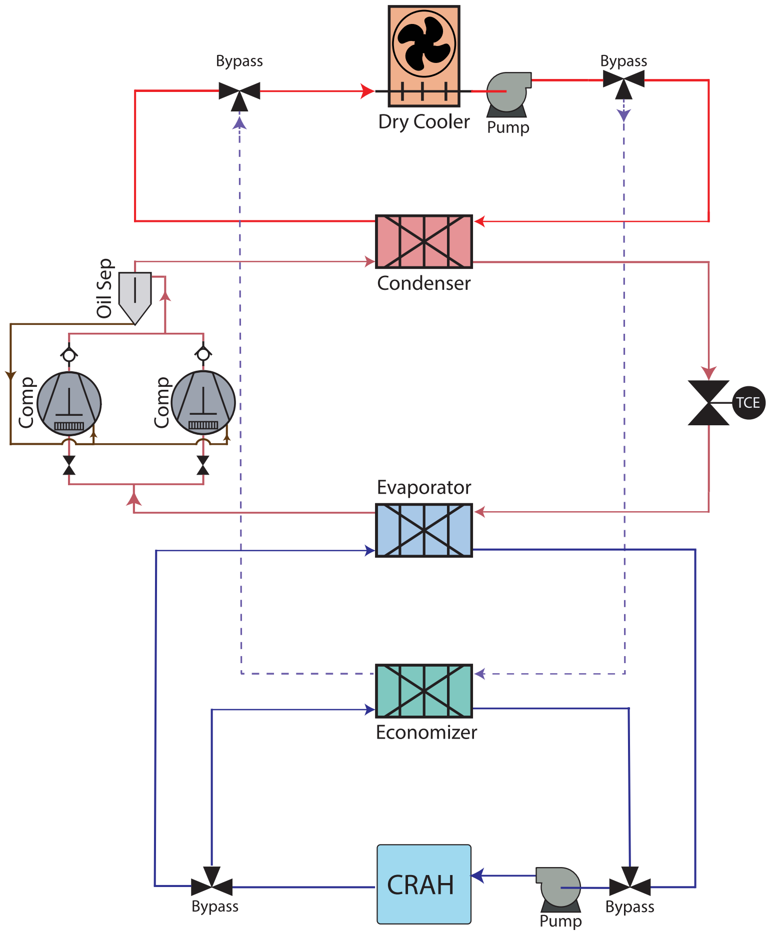
Pumps — glycol (top) and chilled-water (bottom) locations



B&G e-1510 pumps · 2.5AC curve sheet digitized for H(Q) & efficiency → shaft power

Results — Full-Load COP

Full-Load COP — Whole Plant, Every Component Modeled



System P&ID — every block is in the performance model

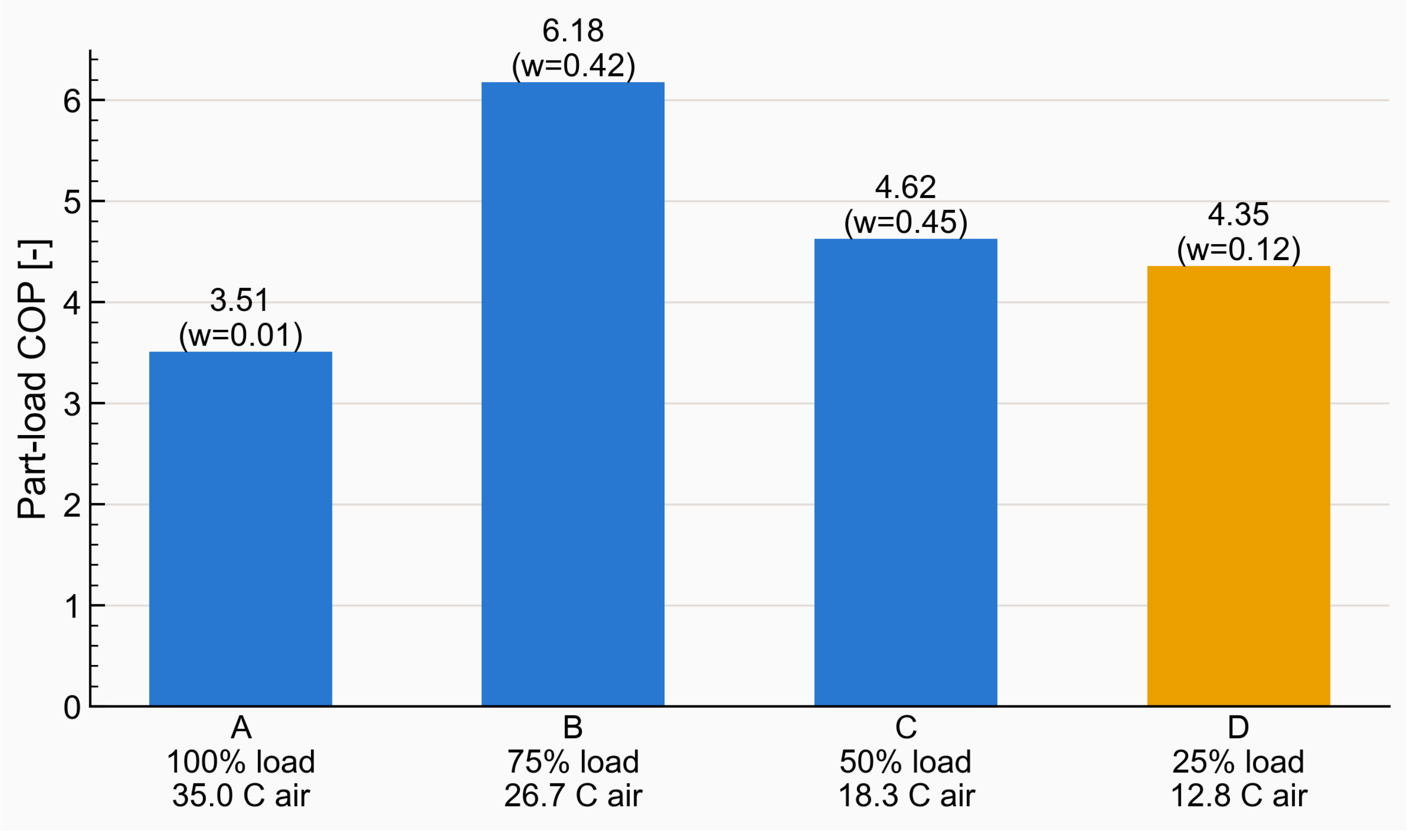
FULL-LOAD COP · WHOLE PLANT · DESIGN POINT

3.59

Every component is modeled, not lumped — compressor (Bitzer AHRI-540 map), dry-cooler EC fans, glycol & chilled-water pumps, economizer and dry cooler — with **all parasitics included** at the design point.

Results — Part Load

IPLV & the 5:1 Turndown



IPLV.IP part-load points A–D per AHRI 550/590-2023

INTEGRATED PART LOAD VALUE

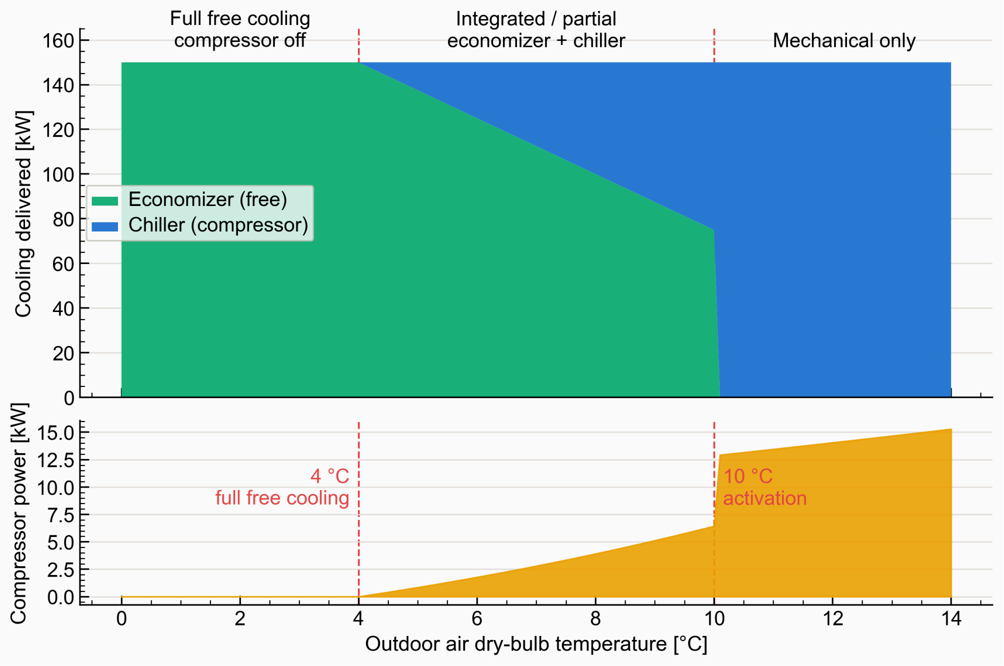
5.23

≥ 5.0 target met with a 4.6% margin. Speed modulation is also what makes the 30 kW minimum load affordable.

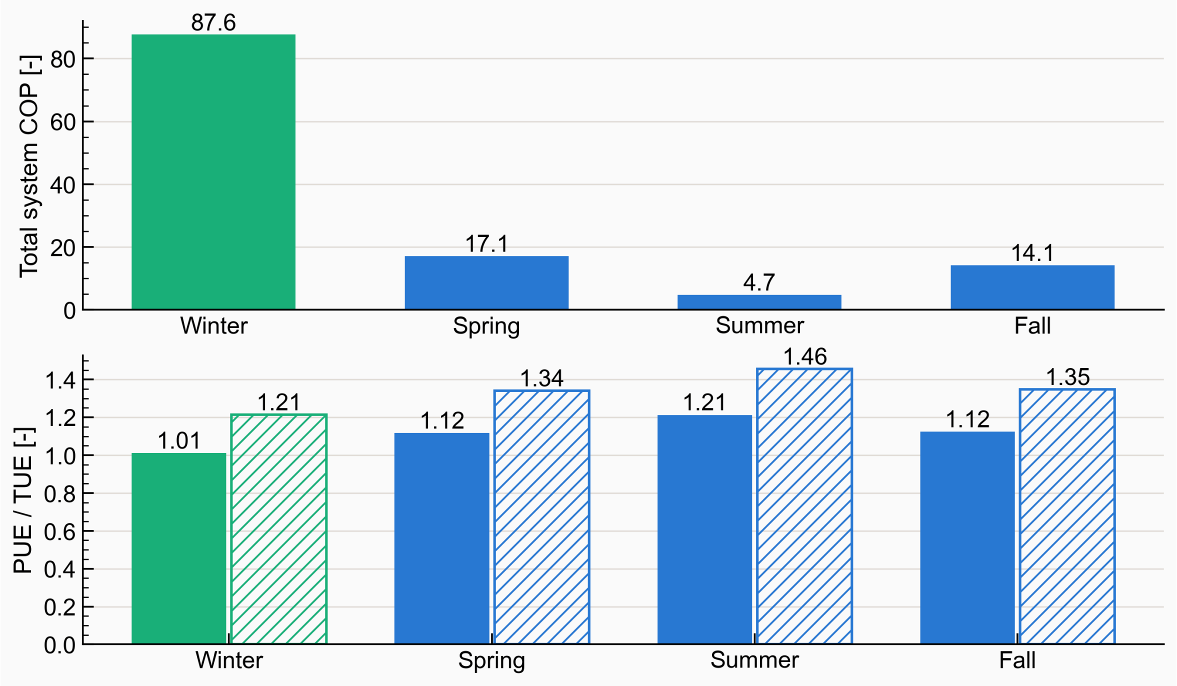
Results — Free Cooling

The Economizer Pays Back Every Winter Hour

Across the economizer band the compressor's share of the load collapses; over the Champaign TMY3 year, **≈40% of all cooling** is delivered with almost no compressor work — winter PUE approaches **1.01**.



Load split & compressor power across the ASHRAE 90.1 economizer envelope

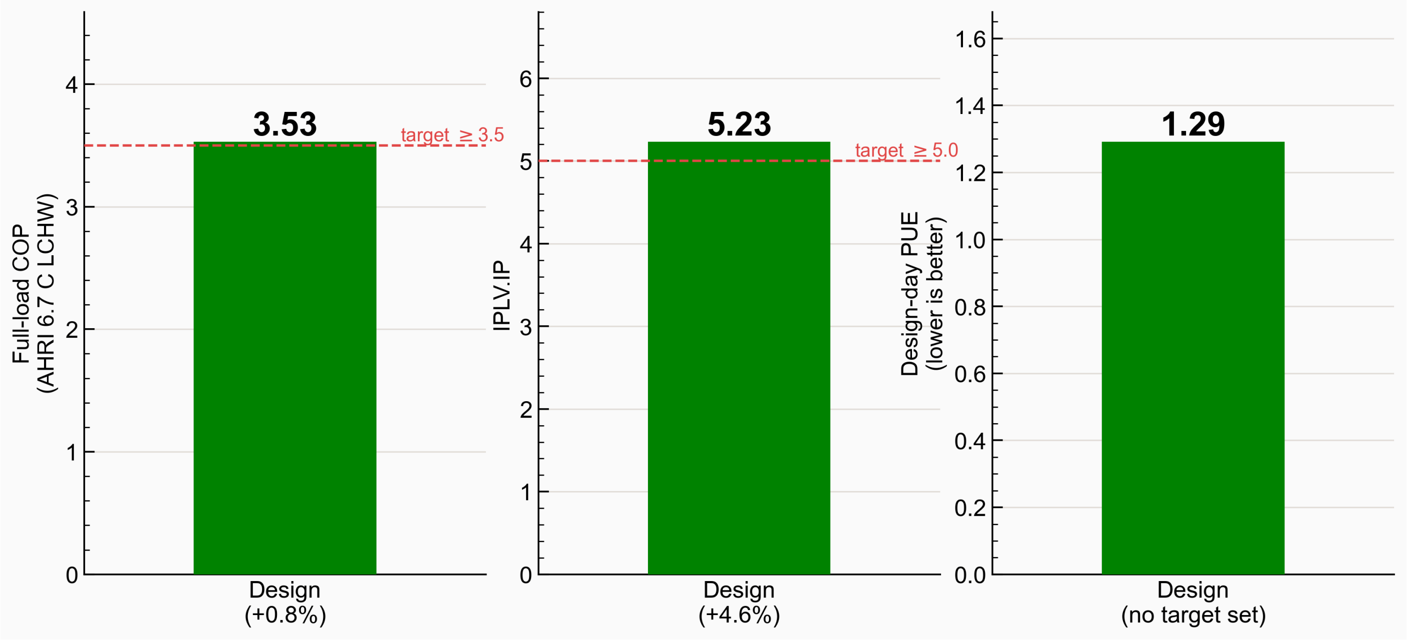


Seasonal-average total-system COP, PUE and TUE

Results — PUE & Compliance

Both Mandatory Targets Cleared

The compressor bank dominates design-day facility power; fans and pumps stay small. The design-day PUE is the worst case — the annual average is far lower.



The design against the two mandatory efficiency targets

DESIGN-DAY FACILITY PUE

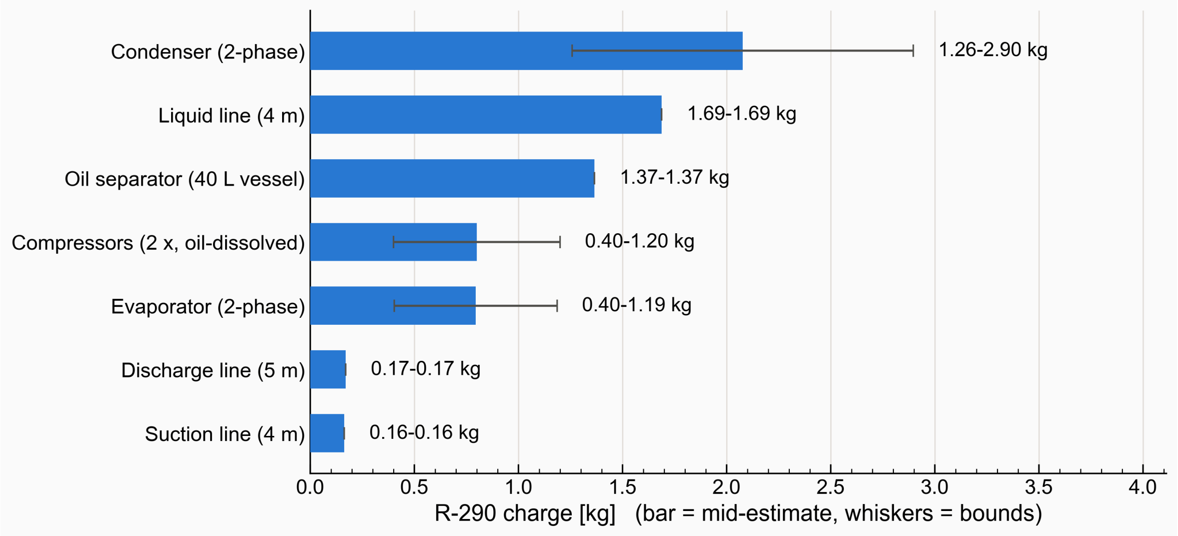
1.29

COP **3.53** ✓ · IPLV **5.23** ✓ — the COP margin is thin, the IPLV margin comfortable. Both computed from the same AHRI-540 map and fan/pump models.

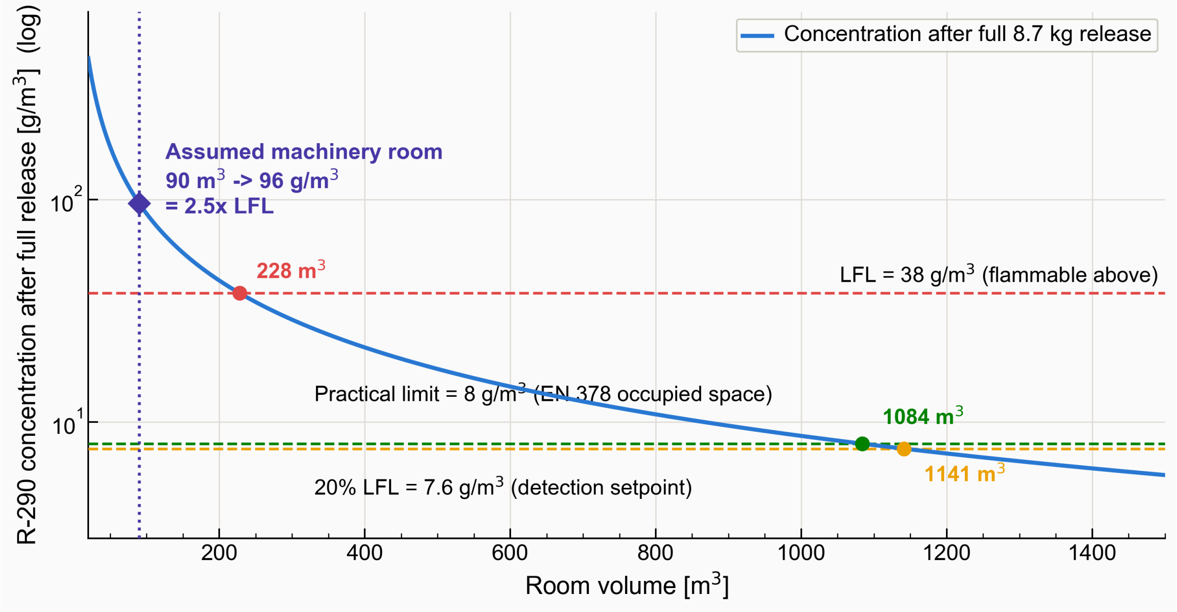
Safety — A3 Charge Analysis

Small Charge, Kept Out of the Hall

- Total R-290 charge is **5.5–8.7 kg** — minimized (no receiver, 4 m liquid line) and kept out of the hall by the glycol loop.
- Dilution alone **won't keep the plant room below LFL** — a full release reaches ~2.5× LFL.
- Safety rests on an **EN 378 machinery room**: 20% LFL gas detection + emergency ventilation.



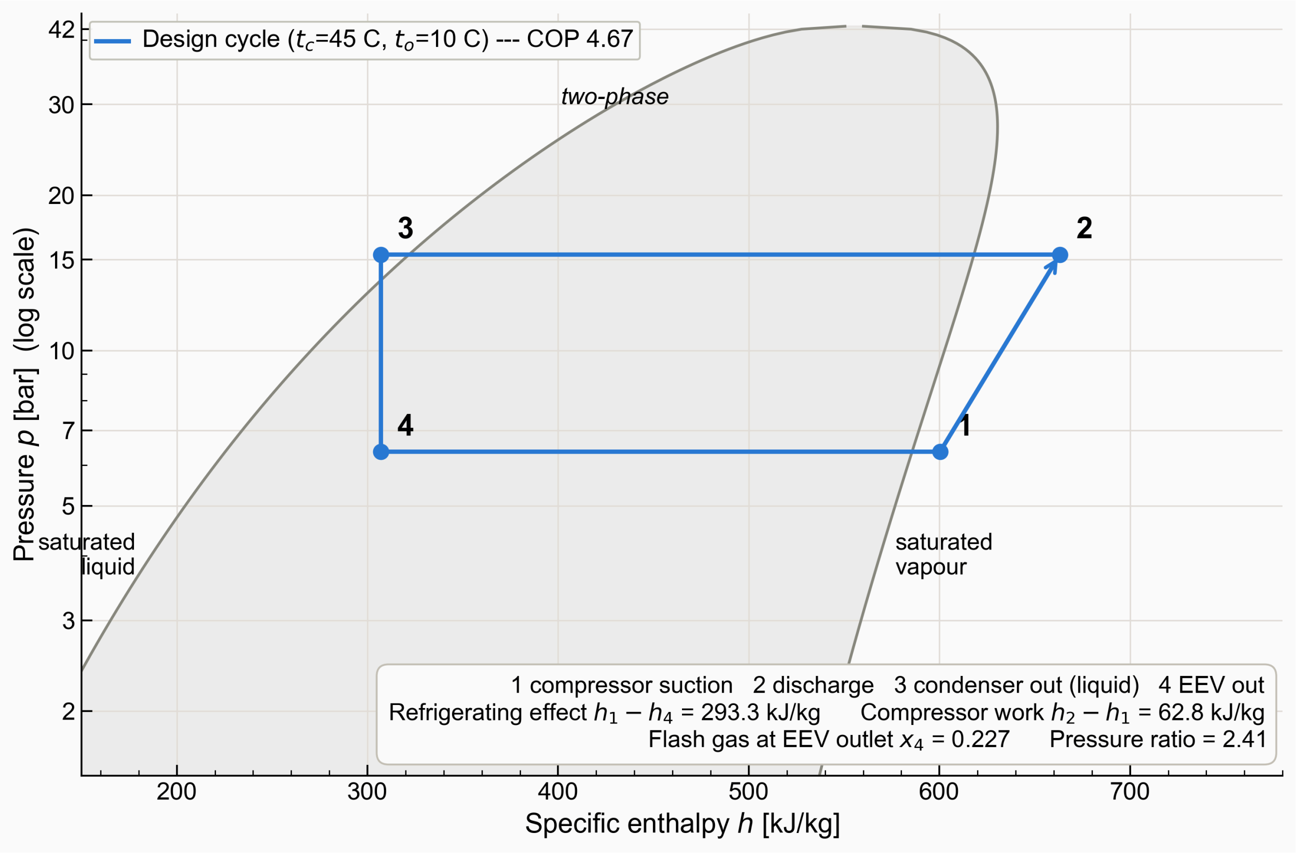
Charge inventory at the design point — condenser holdup dominates



Room concentration after a full release vs. the safety limit

Both Targets Met with an A3-Safe Architecture

- **Both targets met** — COP 3.53 · IPLV 5.23 · PUE 1.29
- Champaign winters deliver **~40% of cooling free** via the economizer
- **R-290 (propane)** working fluid — **low-GWP, A3-safe**



The designed cycle — R-290 p–h chart at 150 kW / 35 °C

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Thank you!

Questions & discussion welcome

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IRCC 2026 · Project #3 — 150 kW R-290 chiller · COP 3.53 · IPLV 5.23 · PUE 1.29